

TECHNICAL PROJECTS REPORT

PROJECT 1: SOILSMART MONITORING SYSTEM

Engineered an IoT smart soil monitoring system using Python microcontrollers, sensors, and Random Forest ML algorithms. Streams real-time soil moisture, pH, and temperature metrics to cloud dashboards, reducing farm water usage by 35%.

PROJECT 2: REAL-TIME EVENT BOOKING & CHAT APP

Developed a cross-platform mobile application using React Native, WebSockets, and Firebase Realtime Database. Handled live seat selection and instant push notifications for over 10,000 active users, optimizing render performance by 35%.

PROJECT 3: SALES ANALYTICS & ETL DASHBOARD

Completed a 3-month Data Analytics internship building executive dashboards in Power BI and writing complex SQL procedures.